

CHE 301 – Chemical Engineering Thermodynamics
Assignment 6

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Abstract

This report presents a comprehensive thermodynamic analysis of carbon dioxide and related systems through three independent but conceptually connected problems.

Problem A focuses on the derivation of a fundamental thermodynamic identity that relates properties such as the thermal expansion coefficient, compressibility factor, and molar volume. The proof establishes the relationship

$$\int_0^P (1 - \beta T) V dP = RT(Z - 1) - \int_0^\rho RT^2 \left(\frac{\partial Z}{\partial T} \right)_\rho \frac{d\rho}{\rho},$$

demonstrating how thermodynamic properties can be linked through the equation of state.

Problem B applies the Peng–Robinson equation of state and the Clausius–Clapeyron relation to evaluate the fugacity of solid carbon dioxide (dry ice) at a high pressure ($P = 100$ atm) and low temperature ($T = -120$ °C). The calculation uses the normal sublimation point ($T_1 = -78.5$ °C, $P_1 = 1$ atm) as a reference, along with thermophysical data for sublimation enthalpy and solid density obtained from *EngineeringToolBox.com*. The results show that at these conditions, the vapor pressure is extremely low and the fugacity of the solid phase increases only slightly with pressure, illustrating the low compressibility of condensed phases.

Problem C investigates the production of dry ice through throttling. Using the Peng–Robinson equation of state to describe both vapor and liquid phases, the mass fraction of dry ice formed during this expansion is calculated based on enthalpy conservation across the valve.

Contents

1 Problem A: Derivation of the Thermodynamic Identity	3
1.1 Objective	3
1.2 Definitions	3
1.3 Use the Definition of β	3
1.4 Express V in Terms of Z and Compute $(\partial V / \partial T)_P$	3
1.5 Substitute and Simplify	4
1.6 Change Variable from P to ρ	4
1.7 Convert Derivative at Constant P to Constant ρ	5
1.8 Boundary Term and Final Identity	5

2	Problem B: Fugacity of Dry Ice at High Pressure	6
2.1	Objective and Given Data	6
2.2	Estimation of Sublimation Pressure Using the Clausius–Clapeyron Relation .	6
2.3	Calculation of Vapor Fugacity Using the Peng–Robinson EOS	7
2.4	Correction of Fugacity from $P_{\text{sat},2}$ to 100 atm	8
2.5	Results Summary	9
2.6	Discussion and Observations	10
3	Problem C: Production of Dry Ice by CO₂ Throttling	11
4	Conclusion	12

1 Problem A: Derivation of the Thermodynamic Identity

1.1 Objective

The objective of this problem is to derive and verify the following thermodynamic identity involving the thermal expansion coefficient and the compressibility factor Z :

$$\int_0^P (1 - \beta T) V dP = RT(Z - 1) - \int_0^P RT^2 \left(\frac{\partial Z}{\partial T} \right)_P \frac{d\rho}{\rho}, \quad Z \equiv \frac{PV}{RT}. \quad (1)$$

1.2 Definitions

- V is molar volume, and $\rho = 1/V$ is molar density.
- R is the gas constant, T is temperature, and P is pressure.
- Thermal expansion coefficient:

$$\beta = \frac{1}{V} \left(\frac{\partial V}{\partial T} \right)_P$$

- Compressibility factor:

$$Z(T, \rho) = \frac{PV}{RT}.$$

1.3 Use the Definition of β

By definition, $\beta V = (\partial V / \partial T)_P$. Hence

$$(1 - \beta T)V = V - T \left(\frac{\partial V}{\partial T} \right)_P,$$

so the integral becomes

$$\int_0^P (1 - \beta T)V dP = \int_0^P V dP - T \int_0^P \left(\frac{\partial V}{\partial T} \right)_P dP. \quad (2)$$

1.4 Express V in Terms of Z and Compute $(\partial V / \partial T)_P$

From $Z = \frac{PV}{RT}$, we get

$$V = \frac{RT}{P} Z(T, \rho).$$

Differentiate at constant P :

$$\left(\frac{\partial V}{\partial T}\right)_P = \frac{R}{P}Z + \frac{RT}{P}\left(\frac{\partial Z}{\partial T}\right)_P = \frac{R}{P}\left[Z + T\left(\frac{\partial Z}{\partial T}\right)_P\right].$$

1.5 Substitute and Simplify

Substitute V and $(\partial V/\partial T)_P$ from Step 2 into Equation (2):

$$\int_0^P (1 - \beta T)V dP = \int_0^P \frac{RT}{P}Z dP - T \int_0^P \frac{R}{P}\left[Z + T\left(\frac{\partial Z}{\partial T}\right)_P\right] dP.$$

Split the second integral:

$$\int_0^P \frac{R}{P}\left[Z + T(\partial Z/\partial T)_P\right] dP = \int_0^P \frac{R}{P}Z dP + \int_0^P \frac{RT}{P}(\partial Z/\partial T)_P dP.$$

Then the integral simplifies:

$$\begin{aligned} \int_0^P (1 - \beta T)V dP &= \underbrace{\int_0^P \frac{RT}{P}Z dP - T \int_0^P \frac{R}{P}Z dP}_{=0} - \int_0^P \frac{RT^2}{P}\left(\frac{\partial Z}{\partial T}\right)_P dP \\ &= - \int_0^P \frac{RT^2}{P}\left(\frac{\partial Z}{\partial T}\right)_P dP. \end{aligned}$$

1.6 Change Variable from P to ρ

Use $P = \rho RTZ(T, \rho)$:

$$dP = RT\left[Z + \rho\left(\frac{\partial Z}{\partial \rho}\right)_T\right]d\rho, \quad \frac{1}{P} = \frac{1}{\rho RTZ}.$$

Substitute into the integral:

$$\begin{aligned} \int_0^P (1 - \beta T)V dP &= - \int_0^\rho \frac{RT^2}{P}\left(\frac{\partial Z}{\partial T}\right)_P dP \\ &= - \int_0^\rho \frac{RT^2}{\rho Z}\left(\frac{\partial Z}{\partial T}\right)_P (Z + \rho Z_\rho) d\rho, \end{aligned}$$

where $Z_\rho \equiv (\partial Z/\partial \rho)_T$.

1.7 Convert Derivative at Constant P to Constant ρ

Use the chain rule:

$$\left(\frac{\partial Z}{\partial T}\right)_P = \left(\frac{\partial Z}{\partial T}\right)_\rho + \left(\frac{\partial Z}{\partial \rho}\right)_T \left(\frac{\partial \rho}{\partial T}\right)_P.$$

Since $\rho = 1/V$:

$$\left(\frac{\partial \rho}{\partial T}\right)_P = -\rho\beta.$$

Thus,

$$\left(\frac{\partial Z}{\partial T}\right)_P = \left(\frac{\partial Z}{\partial T}\right)_\rho - \rho\beta Z_\rho.$$

Substitute into the integral:

$$- \int_0^\rho \frac{RT^2}{\rho Z} [Z_{T|\rho} - \rho\beta Z_\rho] (Z + \rho Z_\rho) d\rho,$$

with $Z_{T|\rho} \equiv (\partial Z / \partial T)_\rho$.

1.8 Boundary Term and Final Identity

Distribute terms and identify the boundary term:

$$I = - \int_0^\rho \frac{RT^2}{\rho} Z_{T|\rho} d\rho + \underbrace{\int_0^\rho \frac{RT^2}{\rho} \rho\beta Z_\rho d\rho}_{RT(Z-1)}.$$

Hence, the final identity is obtained:

$$\boxed{\int_0^P (1 - \beta T) V dP = RT(Z - 1) - \int_0^\rho \frac{RT^2}{\rho} \left(\frac{\partial Z}{\partial T}\right)_\rho d\rho.} \quad (3)$$

2 Problem B: Fugacity of Dry Ice at High Pressure

2.1 Objective and Given Data

The objective of this problem is to determine the fugacity of solid carbon dioxide (dry ice) at a pressure of $P = 100$ atm and a temperature of $T = -120$ °C, using the Peng–Robinson (PR) equation of state to describe vapor-phase properties. The Clausius–Clapeyron equation is employed to estimate the sublimation pressure at the target temperature, and the solid fugacity is related to the vapor fugacity through an exponential pressure correction.

The reference state for sublimation is chosen at $T_1 = -78.5$ °C and $P_1 = 1$ atm, corresponding to the normal sublimation point of CO₂. At this temperature and pressure, solid and vapor phases coexist in equilibrium, providing an appropriate baseline for the Clausius–Clapeyron relation.

The enthalpy of sublimation (ΔH_{sub}) and solid density (ρ_s) were obtained from *EngineeringToolBox.com*. These values are used to evaluate the solid molar volume and to perform the pressure correction for fugacity at elevated pressures.

All necessary data and constants are summarized below:

$$\begin{aligned} T_1 &= -78.5 \text{ °C} = 194.65 \text{ K}, & P_1 &= 1 \text{ atm} = 1.01325 \text{ bar}, \\ T_2 &= -120 \text{ °C} = 153.15 \text{ K}, & P_2 &= 100 \text{ atm} = 101.325 \text{ bar}, \\ R &= 8.314 \text{ kJ}\cdot\text{kmol}^{-1}\text{K}^{-1}, & R &= 0.08314 \text{ m}^3\cdot\text{bar}\cdot\text{kmol}^{-1}\text{K}^{-1}, \\ \Delta H_{\text{sub}} &= 26,000 \text{ kJ}\cdot\text{kmol}^{-1}, & T_c &= 304.2 \text{ K}, \\ P_c &= 73.76 \text{ bar}, & \omega &= 0.225, \\ M &= 44.01 \text{ kg}\cdot\text{kmol}^{-1}, & \rho_s &= 1562 \text{ kg}\cdot\text{m}^{-3}. \end{aligned}$$

2.2 Estimation of Sublimation Pressure Using the Clausius–Clapeyron Relation

The Clausius–Clapeyron equation for sublimation at constant enthalpy is expressed as:

$$\frac{dP}{dT} = \frac{\Delta H_{\text{sub}} P}{RT^2}.$$

Integrating between two states gives:

$$\ln\left(\frac{P_{\text{sat},2}}{P_1}\right) = \frac{\Delta H_{\text{sub}}}{R} \left(\frac{1}{T_1} - \frac{1}{T_2}\right).$$

Substituting the known values:

$$\ln\left(\frac{P_{\text{sat},2}}{1.01325 \text{ bar}}\right) = \frac{26,000 \text{ kJ}\cdot\text{kmol}^{-1}}{8.314 \text{ kJ}\cdot\text{kmol}^{-1}\text{K}^{-1}} \left(\frac{1}{194.65 \text{ K}} - \frac{1}{153.15 \text{ K}} \right) = -4.3535.$$

Thus, the sublimation pressure at $T_2 = -120^\circ\text{C}$ is:

$$P_{\text{sat},2} = (1.01325 \text{ bar}) e^{-4.3535} = 0.0130 \text{ bar}.$$

2.3 Calculation of Vapor Fugacity Using the Peng–Robinson EOS

For the Peng–Robinson (PR) equation of state, the pressure is given by:

$$P = \frac{RT}{V-b} - \frac{a\alpha(T)}{V(V+b) + b(V-b)},$$

where the parameters a , b , κ , and $\alpha(T)$ are defined as:

$$a = 0.45724 \frac{R^2 T_c^2}{P_c}, \quad b = 0.07780 \frac{RT_c}{P_c},$$

$$\kappa = 0.37464 + 1.54226\omega - 0.26992\omega^2, \quad \alpha(T) = \left[1 + \kappa \left(1 - \sqrt{\frac{T}{T_c}} \right) \right]^2.$$

The reduced temperature and pressure are defined as:

$$T_r = \frac{T}{T_c}, \quad P_r = \frac{P}{P_c}.$$

At $T = 153.15 \text{ K}$ and $P = 0.0130 \text{ bar}$, using $T_c = 304.2 \text{ K}$ and $P_c = 73.76 \text{ bar}$, the reduced properties are:

$$T_r = 0.5035, \quad P_r = 1.37 \times 10^{-4}.$$

Using $R = 0.08314 \text{ m}^3\cdot\text{bar}\cdot\text{kmol}^{-1}\text{K}^{-1}$, the calculated constants are:

$$a = 3.9649 (\text{bar}\cdot\text{m}^3\cdot\text{kmol}^{-1})^2, \quad b = 0.02667 \text{ m}^3\cdot\text{kmol}^{-1}.$$

Dimensionless parameters A and B are defined as:

$$A = \frac{a\alpha(T)P}{(RT)^2}, \quad B = \frac{bP}{RT}.$$

At these conditions:

$$A = 4.63 \times 10^{-4}, \quad B = 2.73 \times 10^{-5}.$$

The cubic equation in the compressibility factor Z for the PR EOS is:

$$Z^3 - (1 - B)Z^2 + (A - 3B^2 - 2B)Z - (AB - B^2 - B^3) = 0.$$

The real root corresponding to the vapor phase was determined in Excel using the Goal Seek tool, yielding:

$$Z_{\text{vapor}} = 0.99956.$$

The logarithm of the fugacity coefficient is given by:

$$\ln \phi = (Z - 1) - \ln(Z - B) - \frac{A}{2\sqrt{2}B} \ln \left(\frac{Z + (1 + \sqrt{2})B}{Z + (1 - \sqrt{2})B} \right).$$

Substituting the calculated parameters:

$$\ln \phi = -0.000436, \quad \phi = 0.99956.$$

Hence, the vapor fugacity at the sublimation pressure is:

$$f_v(T_2, P_{\text{sat},2}) = \phi P_{\text{sat},2} = (0.99956)(0.01303 \text{ bar}) = 0.01303 \text{ bar}.$$

2.4 Correction of Fugacity from $P_{\text{sat},2}$ to 100 atm

At equilibrium between the solid and vapor phases:

$$f_s(T_2, P_{\text{sat},2}) = f_v(T_2, P_{\text{sat},2}).$$

The fugacity of the solid at a higher pressure P_2 is obtained using the exponential pressure correction:

$$f_s(T_2, P_2) = f_s(T_2, P_{\text{sat},2}) \exp \left(\frac{\bar{v}_s (P_2 - P_{\text{sat},2})}{R T_2} \right),$$

where $\bar{v}_s$ is the molar volume of the solid, defined as

$$\bar{v}_s = \frac{M}{\rho_s},$$

with M in $\text{kg}\cdot\text{kmol}^{-1}$ and ρ_s in $\text{kg}\cdot\text{m}^{-3}$.

Using the given properties of CO_2 :

$$\bar{v}_s = \frac{44.01 \text{ kg}\cdot\text{kmol}^{-1}}{1562 \text{ kg}\cdot\text{m}^{-3}} = 0.02818 \text{ m}^3/\text{kmol}.$$

The exponential term is evaluated as:

$$\frac{\bar{v}_s(P_2 - P_{\text{sat},2})}{RT_2} = \frac{(0.02818 \text{ m}^3/\text{kmol})[(101.325 - 0.013) \text{ bar}]}{(0.08314 \text{ m}^3\cdot\text{bar}\cdot\text{kmol}^{-1}\text{K}^{-1})(153.15 \text{ K})} = 0.2368.$$

Therefore, the fugacity of the solid at $T_2 = 153.15 \text{ K}$ and $P_2 = 100 \text{ atm}$ is:

$$f_s(T_2, P_2) = (0.01303 \text{ bar}) e^{0.2368} = 0.0163 \text{ bar} = \boxed{0.0161 \text{ atm}}.$$

2.5 Results Summary

All intermediate steps and equations were implemented in **Excel**. The dataset shown in Figure 2 includes all constants and calculated data.

T1, C	-78.5	T1, K	194.65	Tr, K	0.50345168
P1, atm	1	P1, bar	1.01325	Pr, K	1.37362637
T2, C	-120	T2, K	153.15	κ	0.7079838
P2, atm	100	P2, bar	101.325	σ	1.45356455
R, m3bar/kmolK	0.08314	R, kJ/kmolK	8.314	λ_1	0.457235
Tc, C	31.05	Tc, K	304.2	λ_2	0.077796
Pc, atm	72.8	Pc, bar	73.7646	a	3.96487897
ω	0.225			b	0.02667341
M, kg/kmol	44.01			A	0.00046325
ρ , kg/m3	1562			B	2.73E-05
				Z eqn	0.00
				Z	0.99956405
Hsub, kJ/kmol	26000				
ln(Psat@T2/P1)	-4.3535147				
Psat (T2), bar	0.01303194				
ln(ϕ)	-0.00043604				
ϕ	0.99956406				
fv (T2, Psat@T2), bar	0.01302626				
vs, m3/kmol	0.02817542				
fs (T2,P2), bar	0.01629977				
fs (T2,P2), atm	0.01608663				

Figure 1: Excel spreadsheet with the calculations and intermediate parameters used to compute fugacity for CO_2 at $T = -120 \text{ }^\circ\text{C}$ and $P = 100 \text{ atm}$.

A summary of the computed quantities is provided below:

Parameter	Symbol / Equation	Result
Compressibility factor	Z_{vapor}	0.99956
Sublimation pressure at T_2	$P_{\text{sat},2}$	0.0130 bar
Fugacity coefficient (PR)	ϕ	0.99956
Vapor fugacity at $P_{\text{sat},2}$	$f_v = \phi P_{\text{sat},2}$	0.01303 bar
Solid molar volume	$\bar{v}_s = M/\rho_s$	0.02818 m ³ /kmol
Solid fugacity at 100 atm	$f_s = f_v e^{\bar{v}_s(P_2 - P_{\text{sat},2})/(RT_2)}$	0.0163 bar (0.0161 atm)

Table 1: Calculated quantities and corresponding results.

2.6 Discussion and Observations

At -120°C , the vapor pressure of CO_2 is extremely low ($P_{\text{sat}} \approx 0.013$ bar), indicating that sublimation is minimal under these conditions. The Peng–Robinson EOS yields a compressibility factor near one ($Z \approx 1$) and a fugacity coefficient close to one ($\phi \approx 1$), confirming near-ideal vapor behavior at such low pressures.

Because the solid molar volume is very small, increasing external pressure from 1 atm to 100 atm produces only a small increase in solid fugacity—from 0.013 bar to 0.016 bar. This illustrates that the fugacity of condensed phases is far less sensitive to pressure than that of gases.

3 Problem C: Production of Dry Ice by CO₂ Throttling

In this problem, liquid CO at high pressure is expanded through a throttling valve to low pressure, causing part of it to solidify into dry ice. The goal is to determine the mass fraction of CO that becomes solid using enthalpy and the lever rule.

Given Data

State	S1	S2
T (°C)	-35	-78.5
P (bar)	60.795	1.01325

Table 2: States S1 (liquid) and S2 (vapor + solid).

Pressure Conversion

To convert from atm to bar:

$$P \text{ (bar)} = P \text{ (atm)} \times 1.01325$$

$$P_1 = 60 \times 1.01325 = 60.795 \text{ bar}, \quad P_2 = 1 \times 1.01325 = 1.01325 \text{ bar}.$$

Thermodynamic Properties

The compressibility factor Z for each state was found using the `Zcomp(T,P)` function in the CHE calculator:

$$Z_1 = 0.116953112, \quad Z_2 = 0.981401418.$$

The molar enthalpy values were calculated using the `=ENTHALPY(T,P)` function:

$$H_1 = -410599.2858 \text{ kJ kmol}^{-1}, \quad H_v = -397440.0316 \text{ kJ kmol}^{-1}.$$

The enthalpy of sublimation is given as:

$$\Delta H_{\text{sub}} = 26000 \text{ kJ kmol}^{-1}.$$

Lever Rule (Energy Balance)

First, calculate the solid enthalpy at S2:

$$H_s = H_v - \Delta H_{\text{sub}} = -397440.0316 - 26000 = -423440.0316 \text{ kJ kmol}^{-1}.$$

Next, apply the lever rule for the mixture after throttling:

$$x = \frac{H_1 - H_s}{H_v - H_s}$$

$$x = \frac{-410599.2858 - (-423440.0316)}{-397440.0316 - (-423440.0316)} = \frac{12840.7458}{26000} = 0.493874836$$

Thus, the solid (dry ice) mass fraction is:

$$y = 1 - x = 1 - 0.493874836 = 0.506125164.$$

Phase Identification

- State S1 ($T = -35^\circ\text{C}$, $P = 60.795$ bar) — Liquid CO_2
- State S2 ($T = -78.5^\circ\text{C}$, $P = 1.01325$ bar) — Vapor + Solid CO_2

Therefore, after throttling, the CO_2 partially solidifies, forming approximately **50.6% dry ice** and **49.4% vapor**.

	S1 (liquid)	S2 (vapor+solid)
T, C	-35	-78.5
P, bar	60.795	1.01325
Z	0.116953112	0.981401418
H, kJ/kmol	-410599.286	-397440.0316
Hs, kJ/kmol	-423440.032	
x	0.493874836	
y	0.506125164	

Figure 2: Excel spreadsheet with the calculations of mass fraction of dry Ice.

4 Conclusion

The throttling of liquid CO_2 from 60.8 bar and -35°C to 1.01 bar and -78.5°C results in partial solidification due to the Joule–Thomson effect. Using the enthalpy balance and

the lever rule, approximately 50.6% of the CO_2 forms solid dry ice, while 49.4% remains as vapor. This confirms the thermodynamic principle of dry ice production through isenthalpic expansion.